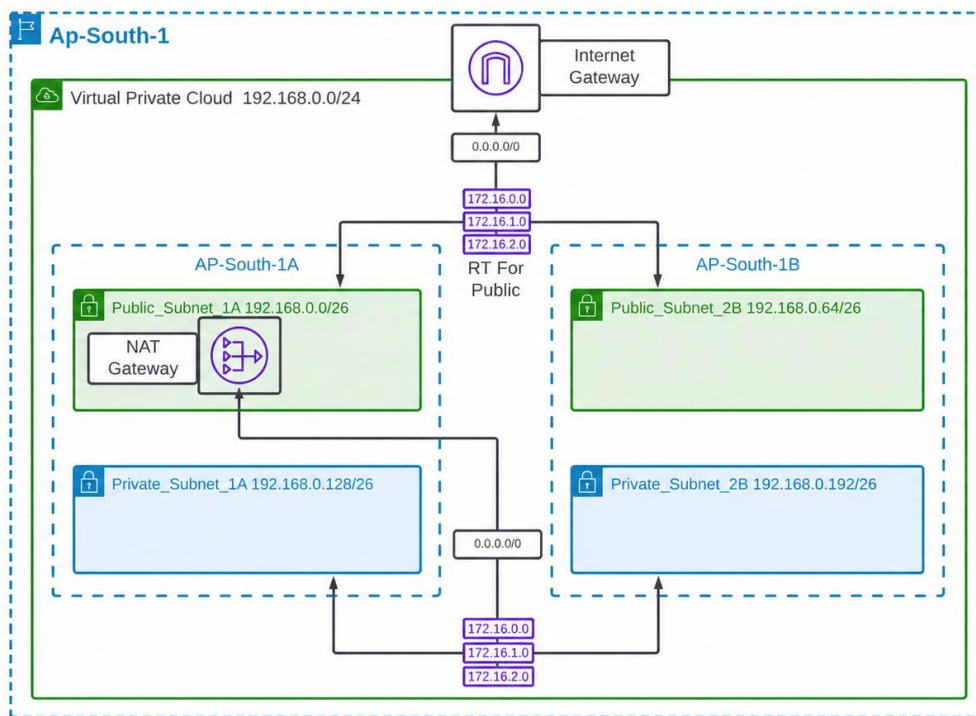


Application Load Balancer Host Based Routing

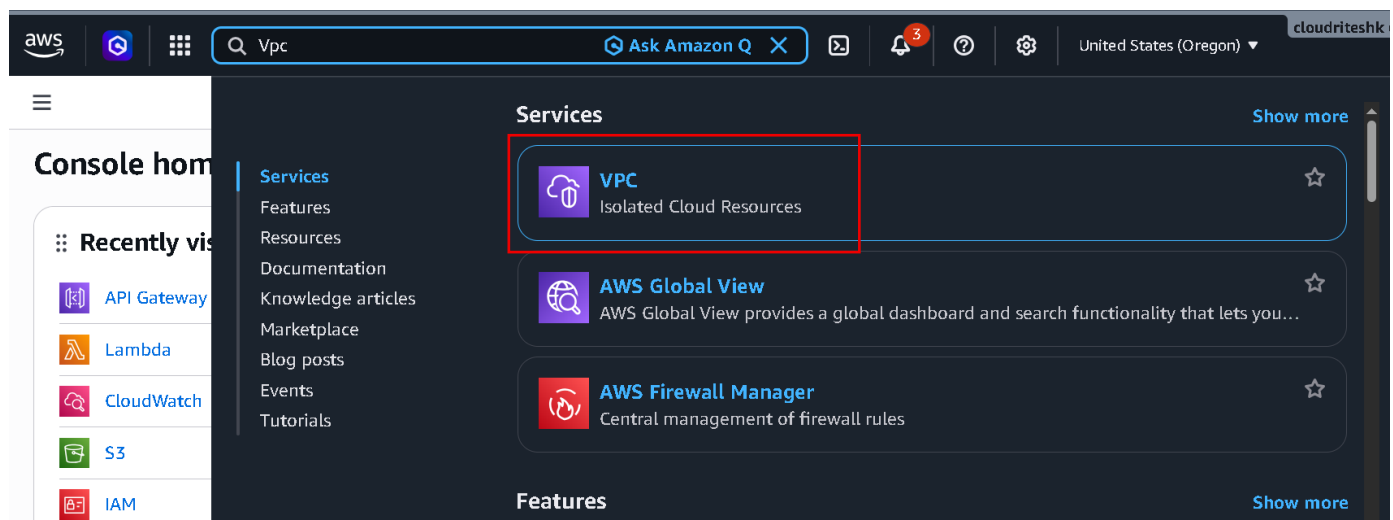
Summary: -

In this I shows how to set up an Application Load Balancer in AWS using host-based routing. Different website requests are sent to different EC2 servers based on the domain name. A custom VPC, public and private subnets, route tables, NAT Gateway, security groups, target groups, EC2 instances, Load Balancer, and Route 53 were configured to make the setup secure and accessible from the internet.

Implementation Load Balancer VPC Design: -



- Firstly, we create our VPC.
- Go to your AWS Console.



- Now Search VPC.
- And open it VPC.

VPC dashboard

AWS Global View

Filter by VPC:

Virtual private cloud

- Your VPCs
- Subnets
- Route tables
- Internet gateways
- Egress-only Internet gateways

Create VPC **Launch EC2 Instances**

Note: Your Instances will launch in the United States region.

Resources by Region Refresh Resources

You are using the following Amazon VPC resources

Resource	Count
VPCs	Oregon 1
Subnets	Oregon 2
Endpoint Services	Oregon 0
NAT Gateways	Oregon 0

- Now simply click Create VPC.

Create VPC [Info](#)

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create [Info](#)

Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional

Creates a tag with a key of 'Name' and a value that you specify.

MyVPC

IPv4 CIDR block [Info](#)

☒ IPv4 CIDR manual input ☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR

192.168.0.0/24

CIDR block size must be between /16 and /28.

- Select VPC only.
- Then Enter your VPC name.
- And then Enter IPv4 CIDR as 192.168.0.0/24

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key **Value - optional**

Q Name X Q MyVPC X Remove tag

Add tag

You can add 49 more tags

Cancel Preview code **Create VPC**

- Now simply scroll down and click Create VPC.
- Now we need to create four Subnet.
- We are creating two public subnets (public subnets having inbound internet access allowed).
- And we create two private subnets (which have no inbound internet access allowed).

☰ VPC > Your VPCs > vpc-03279cdb550dee663 ⓘ 🖨

VPC dashboard <

AWS Global View ↗

Filter by VPC: ▾

▼ Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only Internet gateways

DHCP option sets

vpc-03279cdb550dee663 / MyVPC Actions ▾

Details Info

VPC ID
vpc-03279cdb550dee663

DNS resolution
Enabled

Main network ACL
acl-0f2c2865119107247

IPv6 CIDR (Network border group)
-

State
Available

Tenancy
default

Default VPC
No

Network Address Usage metrics
Disabled

Block Public Access
Off

DHCP option set
dopt-04c9fec64e7acfa5a

IPv4 CIDR
192.168.0.0/24

Route 53 Resolver DNS Firewall rule groups
-

DNS hostnames
Disabled

Main route table
rtb-0457fe28e5ad5cea7

IPv6 pool
-

Owner ID
463646775279

- Now click on subnets.

Subnets (4) Info

Last updated 1 minute ago ⌂ Actions ▾ Create subnet

🔍 Find subnets by attribute or tag

☐	Name ▾	Subnet ID ▾	State ▾	VPC ▾
☐	Subnet1(Public)	subnet-052ee09f56668d6c1	Available	vpc-0862eb08d50c280e6
☐	-	subnet-04ffa8e5e9c79ad3d	Available	vpc-072c4acd761c3b942
☐	Subnet1(Private)	subnet-0508623c58480124a	Available	vpc-0862eb08d50c280e6
☐	-	subnet-0bc545e3030c41ef4	Available	vpc-072c4acd761c3b942

Select a subnet

- Now simply click on Create subnet.
- Firstly, we create two public subnets.

Create subnet Info

VPC

VPC ID
Create subnets in this VPC.

vpc-03279cdb550dee663 (MyVPC) ▾

Associated VPC CIDRs

IPv4 CIDRs
192.168.0.0/24

- Select your VPC that you created now.

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

Public_Subnet_1A

The name can be up to 256 characters long.

Availability Zone [Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

Asia Pacific (Mumbai) / ap-s1-az1 (ap-south-1a)

IPv4 VPC CIDR block [Info](#)

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

192.168.0.0/24

IPv4 subnet CIDR block

192.168.0.0/26

64 IPs

- Enter Subnet name as Public_Subnet_1A
- The select Availability Zone (ap-south-1a).
- Then IPv4 subnet CIDR block as 192.168.0.0/26
- Now simply scroll down.

▼ Tags - optional

Key

Q Name



Value - optional

Q Public_Subnet_1A



Remove

Add new tag

You can add 49 more tags.

Remove

Add new subnet

- Now we are going to create another click Add new Subnet.

Subnet 2 of 2

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

Public_Subnet_2B

The name can be up to 256 characters long.

Availability Zone [Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

Asia Pacific (Mumbai) / ap-s1-az3 (ap-south-1b)

IPv4 VPC CIDR block [Info](#)

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

192.168.0.0/24

IPv4 subnet CIDR block

192.168.0.64/26

64 IPs

- Enter subnet 2 name as Public_Subnet_2B
- Select Availability Zone (ap-south-1b)
- Then Enter IPv4 subnet CIDR block as 192.168.0.64/26

- Now simply scroll down.

▼ Tags - optional

Key	Value - optional
Name	Public_Subnet_1A

[Add new tag](#)
You can add 49 more tags.

[Remove](#)

[Add new subnet](#)

- Now click Add new subnet.
- Now we will add Private Subnets.

Subnet 3 of 3

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

IPv4 subnet CIDR block
 64 IPs
[<](#) [>](#) [^](#) [v](#)

- Enter Subnet name as Private _Subnet_ 1A
- The select Availability Zone (ap-south-1a).
- Then IPv4 subnet CIDR block as 192.168.0.128/26
- Now simply scroll down.

▼ Tags - optional

Key	Value - optional
Name	Public_Subnet_1A

[Add new tag](#)
You can add 49 more tags.

[Remove](#)

[Add new subnet](#)

- Now we are going to create another click Add new Subnet.

Subnet 4 of 4

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

Private_Subnet_2B

The name can be up to 256 characters long.

Availability Zone [Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

Asia Pacific (Mumbai) / ap-south-1b

IPv4 VPC CIDR block [Info](#)

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

192.168.0.0/24

IPv4 subnet CIDR block

192.168.0.192/26

64 IPs

- Enter Subnet name as Private_Subnet_2B
- The select Availability Zone (ap-south-1b).
- Then IPv4 subnet CIDR block as 192.168.0.192/26
- Now simply scroll down.

▼ Tags - optional

Key

Q Name

Value - optional

Q Private_Subnet_2B

Remove

Add new tag

You can add 49 more tags.

Remove

Add new subnet

Cancel

Create subnet

- Now click Create subnet.

▼ Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only Internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Subnets (4) [Info](#)

Last updated
5 minutes ago

Actions

Create subnet

Find subnets by attribute or tag

Subnet ID : subnet-0e7823bb0276394a2

Subnet ID : subnet-0eeb1e9dd70507728

Subnet ID : subnet-0f82985187f3f7b70

(+1)

Clear filters

☐	Name	Subnet ID	State	VPC
☐	Public_Subnet_1A	subnet-0e7823bb0276394a2	Available	vpc-03279cdb550dee663 M
☐	Private_Subnet_2B	subnet-0c053b6393c3e8c5f	Available	vpc-03279cdb550dee663 M
☐	Private_Subnet_1A	subnet-0f82985187f3f7b70	Available	vpc-03279cdb550dee663 M
☐	Public_Subnet_2B	subnet-0eeb1e9dd70507728	Available	vpc-03279cdb550dee663 M

- Now our Four Subnet is created.
- Now to provide Internet connectivity, we have to add Internet gateway and we have to attach internet gateway with our VPC.
- Now click Internet gateways.

Internet gateways (2) [Info](#)

Find internet gateways by attribute or tag

Actions

Create internet gateway

Name Internet gateway ID State VPC ID

- Now click Create internet gateway.

Create internet gateway [Info](#)

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag

Creates a tag with a key of 'Name' and a value that you specify.

IG

- Now Enter Name (IG).

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Name

Value - optional

IG

Remove

Add new tag

You can add 49 more tags.

Cancel

Create internet gateway

- Now simply click Create internet gateway.

igw-050a2613d0c562441 / IG

Details [Info](#)

Internet gateway ID

igw-050a2613d0c562441

State

Detached

VPC ID

-

Owner

4636

Actions

Attach to VPC

Detach from VPC

Manage tags

Delete

Tags (1)

Search tags

Key Value

Name IG

Manage tags

- Now our internet gateway is created.
- But you see it's not Attach with our VPC.
- Now click Action and then select Attach to VPC.

Attach to VPC (igw-050a2613d0c562441) [Info](#)

VPC

Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs

Attach the internet gateway to this VPC.

vpc-03279cdb550dee663

AWS Command Line Interface command

Cancel

Attach internet gateway

- Now Select Create VPC and then Click Attach internet gateway.

Virtual private cloud > Internet gateways > igw-050a2613d0c562441

igw-050a2613d0c562441 / IG

Details Info
 Internet gateway ID
 igw-050a2613d0c562441
 State
 Attached
 VPC ID
 vpc-03279cdb550dee663 | MyVPC
 Owner
 463646775279

Tags (1) Manage tags
 Search tags
 < 1 >

Key	Value
Name	IG

- it's Attach to VPC.
- Now we have to set up route table.
- Now we have a main route table which is default and we want to provide inbound connectivity to this public subnet.
- So, for this I am going to create Separate Route table for public subnet
- Now click on Route tables in nav bar.

Route tables (1/5)

Find route tables by attribute or tag

Last updated 1 minute ago

Create route table

Name	Route table ID	Explicit subnet associ...	Edge associations
Main	0457fe28e5ad5cea7	-	-
	017299725a38d5666	2 subnets	-
	0aee4d9128c0e3ce0	-	-
vpc1-	0e19b34caf5643c37	subnet-0508623c584801...	-
vpc1-	0bad251f343575c0f	subnet-052ee09f56668d...	-

- Now 1st one is our Default Route table we name it as Main and save it.
- we know that this is our VPC default Route table by clicking Route table ID.
- Now click on Create route table.

Create route table

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
 Create a tag with a key of 'Name' and a value that you specify.
 RT For Public

VPC
 The VPC to use for this route table.
 vpc-03279cdb550dee663 (MyVPC)

- Enter name of Route table (RT For Public)
- Now select VPC that you created now.

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key	Value - optional
Name	RT For Public

Add new tag

You can add 49 more tags.

Create route table

- Now simply scroll down and click Create route table.

Virtual private cloud

- Your VPCs
- Subnets
- Route tables**
- Internet gateways
- Egress-only Internet gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways

rtb-0b72240aa9b00dfea / RT For Public
Actions

Details Info

Route table ID rtb-0b72240aa9b00dfea	Main No	Explicit subnet associations -	Edge associations -
VPC vpc-03279cdb550dee663 MyVPC	Owner ID 463646775279		

Routes
Subnet associations
Edge associations
Route propagation
Tags

- Now our route table is created.
- But right now, your route table is not associated with public subnets.
- Now click on Route table.

Route tables (1/6) Info Last updated 13 minutes ago Actions Create route table

Find route tables by attribute or tag

	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☐	Main	rtb-0457fe28e5ad5cea7	-	-	Yes	vpc-03279cdb550
☒	RT For Public	rtb-0b72240aa9b00dfea	-	-	No	vpc-03279cdb550
☐	-	rtb-017299725a38d5666	2 subnets	-	Yes	vpc-072c4acd761
☐	-	rtb-0aee4d9128c0e3ce0	-	-	Yes	vpc-0862eb08d5c

rtb-0b72240aa9b00dfea / RT For Public
Details
Routes
Subnet associations
Edge associations
Route propagation
Tags

Explicit subnet associations (0) Edit subnet associations

- Now select RT For Public route table.
- And click Subnet associations.
- And then click Edit subnet association.

Available subnets (2/4) Filter subnet associations 1

	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☒	Public_Subnet_1A	subnet-0e7823bb0276394...	192.168.0.0/26	-	Main (rtb-0457fe28e5ad5cea7 /
☐	Private_Subnet_2B	subnet-0c053b6393c3e8c5f	192.168.0.192/26	-	Main (rtb-0457fe28e5ad5cea7 /
☐	Private_Subnet_1A	subnet-0f82985187f3f7b70	192.168.0.128/26	-	Main (rtb-0457fe28e5ad5cea7 /
☒	Public_Subnet_2B	subnet-0eeb1e9dd705077...	192.168.0.64/26	-	Main (rtb-0457fe28e5ad5cea7 /

Selected subnets

[subnet-0eeb1e9dd70507728 / Public_Subnet_2B](#)
[subnet-0e7823bb0276394a2 / Public_Subnet_1A](#)

Cancel
Save associations

- Now select both public Subnets.
- And click Save association.
- Now we have to add this particular route towards the internet gateway into the route public.

✓ You have successfully updated subnet associations for rtb-0b72240aa9b00dfea / RT For Public. ×

Route tables (1/6) [Info](#) Last updated 3 minutes ago [Actions](#) [Create route table](#)

Find route tables by attribute or tag

	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☐	Main	rtb-0457fe28e5ad5cea7	–	–	Yes	vpc-03279cdb550
☒	RT For Public	rtb-0b72240aa9b00dfea	2 subnets	–	No	vpc-03279cdb550
☐	–	rtb-017299725a38d5666	2 subnets	–	Yes	vpc-072c4acd761

rtb-0b72240aa9b00dfea / RT For Public ⚙️ ▼

[Details](#) [Routes](#) [Subnet associations](#) [Edge associations](#) [Route propagation](#) [Tags](#)

Routes (1) Both [Edit routes](#)

Filter routes

Destination	Target	Status	Propagated	Route Origin

- Now click Routes and then click Edit routes.

Route 2

Destination: ×

Target: ▼

×

Status: –

Propagated: No

Route Origin: CreateRoute

[Remove](#)

[Add route](#)

[Cancel](#) [Preview](#) [Save changes](#)

- Now Firstly, click Add route and then Enter 2nd Route details.
- Select Destination 0.0.0.0/0
- Select Target Internet Gateway
- Then select Created Internet Gateway
- And simply click Save Changes.
- Now this is done and we need to Setup NAT Gateway.
- Because you will place your Ec2 instance into the private subnet.
- If we are not going to set up NAT Gateway then you cannot provide outbound internet access to your EC2 instance.
- It means you cannot update your Ec2 instance.
- So, we have to create this NAT Gateway inside the public subnet.

[VPC](#) > [Route tables](#) > rtb-0b72240aa9b00dfea ⓘ 🔍

▼ Virtual private cloud

- Your VPCs
- Subnets
- [Route tables](#)
- Internet gateways
- Egress-only Internet gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- [NAT gateways](#)

rtb-0b72240aa9b00dfea / RT For Public [Actions](#)

Details [Info](#)

Route table ID rtb-0b72240aa9b00dfea	Main ☒ No	Explicit subnet associations 2 subnets	Edge associations –
VPC vpc-03279cdb550dee663 MyVPC	Owner ID 463646775279		

[Routes](#) [Subnet associations](#) [Edge associations](#) [Route propagation](#) [Tags](#)

- Now click NAT Gateway in nav bar.

NAT gateways (1) [Info](#)

Find NAT gateways by attribute or tag

< 1 > ⚙️

	Name	NAT gateway ID	Connectivity...	State	State message
☐	vpc1-nat	nat-0c45ca4968221ec7a	Public	Available	—

- Now click Create NAT gateway.

Create NAT gateway [Info](#)

A highly available, managed Network Address Translation (NAT) service that instances in private subnets can use to connect to services in other VPCs, on-premises networks, or the internet.

NAT gateway settings

Name - *optional*

Create a tag with a key of 'Name' and a value that you specify.

NAT

The name can be up to 256 characters long.

Availability mode [Info](#)

Choose whether to deploy across all zones in the region or restrict to a single availability zone.

☐ Regional - *new*

Scales automatically across all regional AZs, simplifying management for multi AZ deployments.

☒ Zonal

Provides granular control within a specific availability zone, adhering to subnet level settings.

Subnet

Select a subnet in which to create the NAT gateway.

subnet-0e7823bb0276394a2 (Public_Subnet_1A)

- Now Enter Gateway name.
- Availability mode as Zonal.
- Select Public_Subnet_1A
- Now scroll down.

Connectivity type

Select a connectivity type for the NAT gateway.

☒ Public

☐ Private

Elastic IP allocation ID [Info](#)

Assign an Elastic IP address to the NAT gateway.

eipalloc-08b23e8abf29d11fb

Allocate Elastic IP

Additional settings [Info](#)

- Now click Allocate Elastic IP.
- Now scroll down.

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name

Value - *optional*

Q NAT

Remove

Add new tag

You can add 49 more tags.

Cancel

Create NAT gateway

- Now click Create NAT gateway.
- Now we have to setup this 0.0 towards the net gateway.

▼ Virtual private cloud

- Your VPCs
- Subnets
- Route tables**
- Internet gateways
- Egress-only Internet gateways
- DHCP option sets

NAT gateways (2) [Info](#)

Find NAT gateways by attribute or tag

	Name	NAT gateway ID	Connectivity...	State	State message
☐	vpc1-nat	nat-0c45ca4968221ec7a	Public	✓ Available	–
☐	NAT	nat-0b0315b1af281d5f4	Public	✓ Available	–

- Our NAT gateway is created.
- Now we have need to connect the NAT Gateway to our Main Route table.
- For we go inside route table for that click Route tables.

Route tables (1/6) [Info](#)

Last updated 34 minutes ago

Find route tables by attribute or tag

	Name	Route table ID	Explicit subnet associ...	Edge associations
☒	Main	rtb-0457fe28e5ad5cea7	–	–
☐	RT For Public	rtb-0b72240aa9b00dfea	2 subnets	–
☐	–	rtb-017299725a38d5666	2 subnets	–
☐	–	rtb-0aee4d9128c0e3ce0	–	–

rtb-0457fe28e5ad5cea7 / Main

Details **Routes** Subnet associations Edge associations Route propagation Tags

Routes (1)

Filter routes

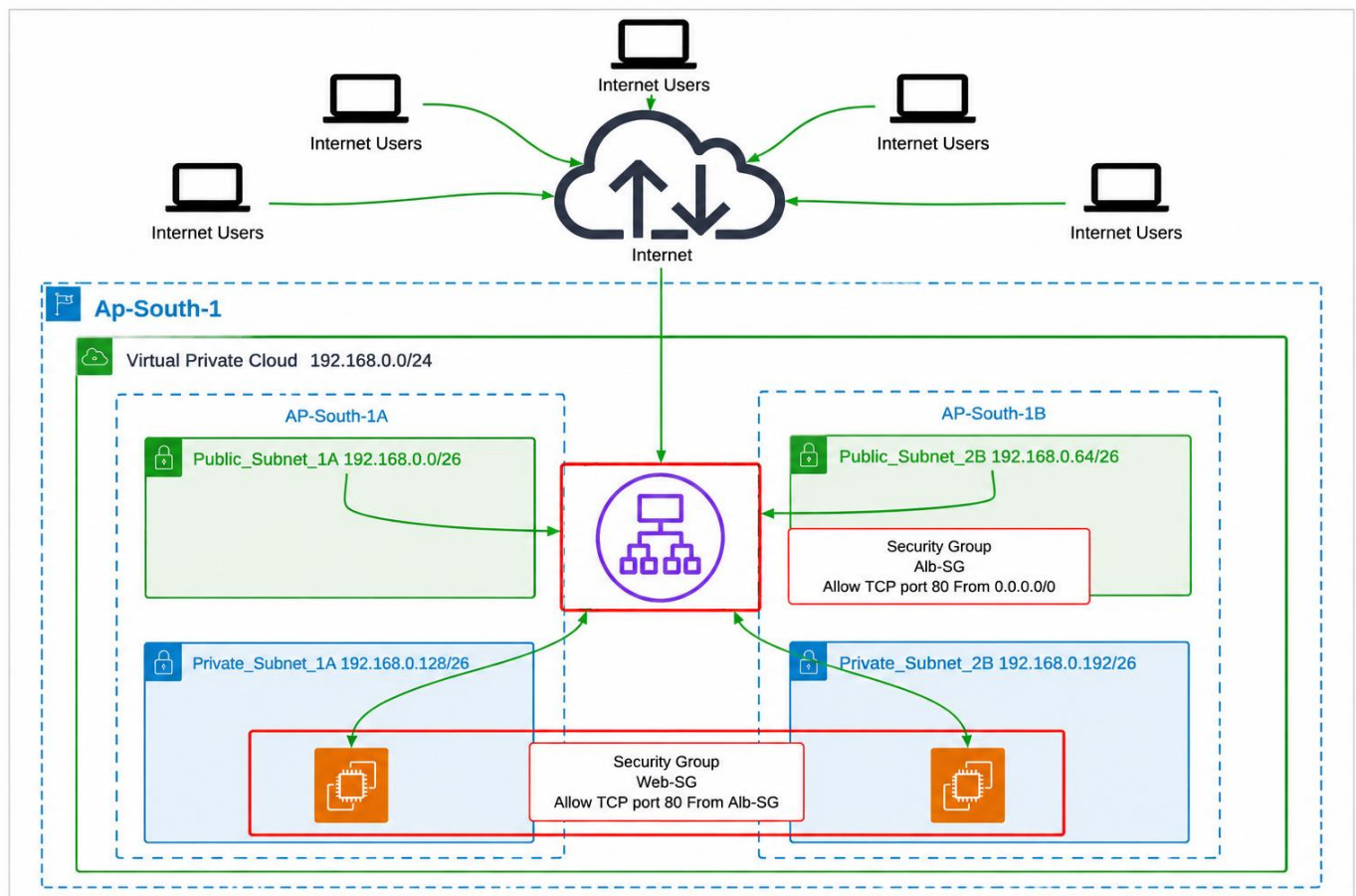
Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	NAT Gateway nat-0b0315b1af281d5f4	–	No	CreateRoute

[Add route](#) [Remove](#)

Cancel [Preview](#) [Save changes](#)

- Now Firstly, click Add route and then Enter 2nd Route details.
- Select Destination 0.0.0.0/0
- Select Target NAT Gateway
- Then select Created NAT Gateway
- And simply click Save Changes.
- And now we are ready to implement load balancer.
- As per rule when you are going to setup your load balancer you have to follow VPC best practices.
- And whatever we are doing right now we are setting up this according to the AWS best practices.

Now Setup Security Group: -



- So, in this image you see users from all over the internet will send requests to my load balancer URL.
- And then my load balancer will send this request to the EC2 instance.
- We have to setup security group at two level.
- The first level we have is to set up a security group for load balancers(Alb-SG).
- And as we know that users over the internet will send requests to your load balancers on a port number 80.
- So, I just created other security group web-SG allows TCP port no 80 from(Alb-SG) anywhere because anybody over the internet can open website.
- And Second level we set up security group for the source.
- Because accept the request if the request is coming from only load balancer.



▼ Security

Network ACLs

Security groups

▼ PrivateLink and Lattice

Getting started

Endpoints

Endpoint services

Service networks

Create VPC

Launch EC2 Instances

Note: Your Instances will launch in the Asia Pacific region.

Resources by Region

You are using the following Amazon VPC resources

VPCs

Mumbai 3

► See all regions

- Now opens VPC and in Nav bar scroll down.

- And click Security groups in Security.

Security Groups (1444) [Info](#) [Actions](#) [Export security groups to CSV](#) [Create security group](#)

< 1 2 3 4 5 6 7 ... 29 > [Settings](#)

☐	Name	Security group ID	Security group name	VPC ID
☐	-	sg-00d8665dd4d518218	launch-wizard-1	vpc-072c4acd7
☐	-	sg-02160387bc543e3b4	databasepostgres1sg1	vpc-072c4acd7
☐	-	sg-080496f1c85d4e152	launch-wizard-242	vpc-072c4acd7
☐	-	sg-0e3aff4580fec321a	launch-wizard-772	vpc-072c4acd7

Select a security group

- Click security groups.

Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)

Name cannot be edited after creation.

Description [Info](#)

VPC [Info](#)

- Enter security group name (Alb-SG).
- Then Enter description (Securing My ALB From Internet).
- And then select Create VPC.

Inbound rules [Info](#)

This security group has no inbound rules.

[Add rule](#)

- Now click Add rule.

Inbound rules [Info](#)

Type [Info](#)

Protocol

Port range [Info](#)

Source [Info](#)

Description - optional [Info](#)

[Info](#)

[Delete](#)
[Add rule](#)

- Now select type as HTTP.
- And Select source From Anywhere.
- Why anywhere because we are going to get request from internet so that anybody can open my website.

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tags

[Cancel](#)

[Create security group](#)

- Now simply scroll down and click Create Security group.

Security Groups (1) [Info](#)

[Actions](#) [Export security groups to CSV](#) [Create security group](#)

[Alb-SG](#) [Clear filters](#)

☐

Name	Security group ID	Security group name	VPC ID
-	sg-0b3ee61463ef94613	Alb-SG	vpc-03279cdb55

Select a security group

- First, Security group is created.
- Now we create 2nd Security group.
- So, click create security group.

Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)

web-SG

Name cannot be edited after creation.

Description [Info](#)

Securing my web servers and allowing traffic from ALB only

VPC [Info](#)

vpc-03279cdb550dee663 (MyVPC)

- Enter Security group name (web-SG)
- Enter Description(Securing my web servers and allowing traffic from ALB only).
- Select your Created VPC.

Inbound rules [Info](#)

Security group rule ID	Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info	
sg-088aa13bef1d01cc0	HTTP	TCP	80	Cu...		Delete
-	SSH	TCP	22	An...	0.0.0.0/0	Delete

[Add rule](#)

- Now add Inbound rules.
- Select type HTTP.
- Then Select Source Alb-SG security group that you created.

- And then add one more inbound rules.
- Select type SSH
- And then source Anywhere.

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

[Add new tag](#)

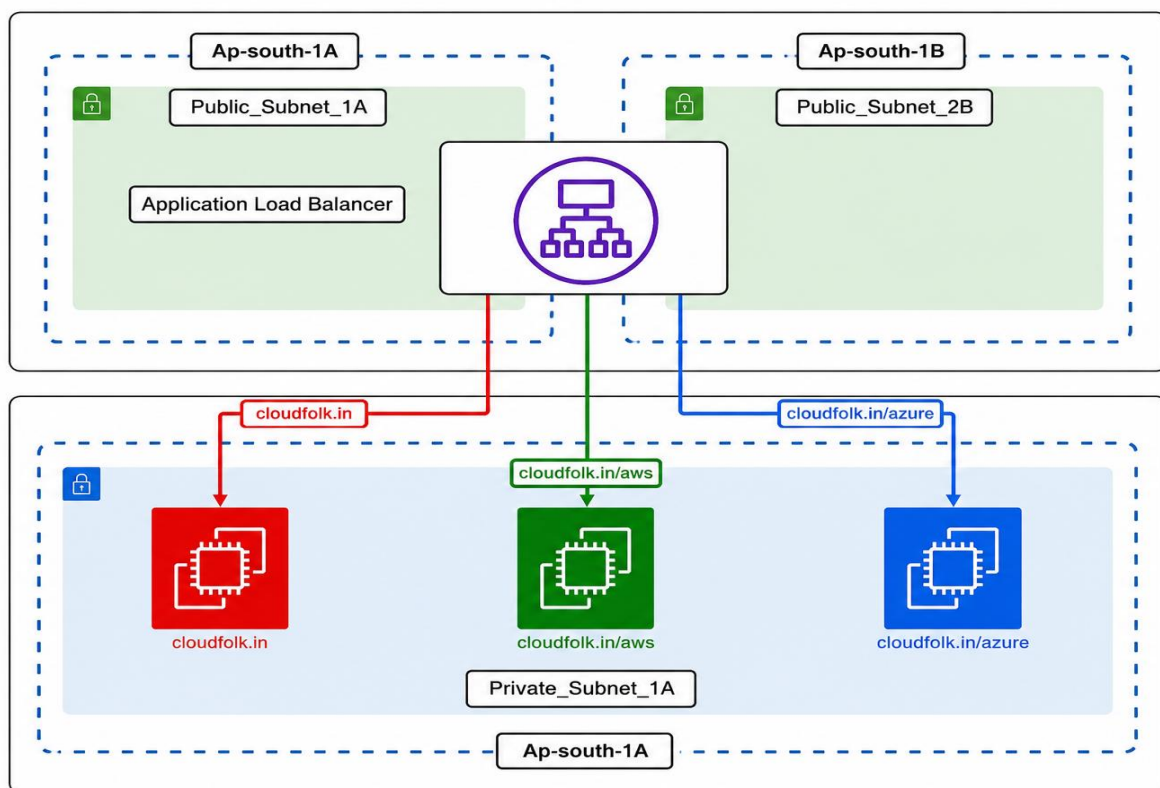
You can add up to 50 more tags

[Cancel](#)

[Create security group](#)

- Now simply scroll down and click Create Security group.
- Now our two Security group is ready.
- This is the basic thing we need in implementing host based routing.

Now Let's implement host-based routing: -



- Our main aim is implementing host-based routing.
- So that I created one private_subnet_1A and I have only one EC2 instance in each target.
- So, first thing is that we have to set up three targets and we have to create three EC2 instances.
- We have to set up the load balancer as well.
- And also, we need to create two security groups one for load balancer and one for our EC2 instances.
- So, start with creating two security groups.
- Previously we created our security groups we use that security groups (Alb-SG and web-SG).
- Now we need to create three Ec2 instance cloudfolk.in, cloudfolk.in/aws, cloudfolk.in/azure.
- And when you create in network setting use your created VPC and all EC2 in subset Private_Subset_1A .
- And choose security group web-SG.

- Let's create our Three EC2 instance.

The screenshot shows the AWS Management Console for EC2 Instances. On the left is a navigation menu with 'Instances' selected. The main area shows 'Instances (2)' with a table of running instances. The 'Launch instances' button is highlighted with a red box. Below the table, there's a 'Select an instance' section.

Name	Instance ID	Instance state	Instance type	Status check	Alarm status
Private-Instance	i-002f9967d6bbb860e	Running	t2.medium	2/2 checks passed	View alarm
Public-Instance1	i-0c0c972f67af41ed2	Running	t2.medium	2/2 checks passed	View alarm

- Now opens EC2 in new tab.
- And click Launch instance.

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

[Add additional tags](#)

- Enter instance name (web-server-1).
- Scroll down.

Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE

[Browse more AMIs](#)

Including AMIs from AWS, Marketplace and the Community

- Choose OS in my case I choose Amazon Linux.
- Scroll down

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t3.micro

Free tier eligible

Family: t3 2 vCPU 1 GiB Memory Current generation: true

On-Demand SUSE base pricing: 0.0104 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour

On-Demand RHEL base pricing: 0.0392 USD per Hour

On-Demand Windows base pricing: 0.0196 USD per Hour

On-Demand Linux base pricing: 0.0104 USD per Hour

☐ All generations

[Compare instance types](#)

[Additional costs apply for AMIs with pre-installed software](#)

- Choose t3.micro this is Free tier eligible.
- Scroll down.

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

Select

[Create new key pair](#)

- Click Create new key pair.

Create key pair



Key pair name

Key pairs allow you to connect to your instance securely.

demo

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ RSA

RSA encrypted private and public key pair

☐ ED25519

ED25519 encrypted private and public key pair

Private key file format

☒ .pem

For use with OpenSSH

☐ .ppk

For use with PuTTY

[Cancel](#)

[Create key pair](#)

- Enter key name.
- Key pair type as RSA
- And .pem file
- And last click Create key pair
- Now your keypair is downloaded and save it.
- Now scroll down.

▼ **Network settings** [Info](#)

Edit

Network | [Info](#)
vpc-072c4acd761c3b942

Subnet | [Info](#)
No preference (Default subnet in any availability zone)

Auto-assign public IP | [Info](#)
Enable
Additional charges apply when outside of [free tier allowance](#)

- Now simple Edit.

▼ **Network settings** [Info](#)

VPC - required | [Info](#)

vpc-03279cdb550dee663 (MyVPC)
192.168.0.0/24

Subnet | [Info](#)

subnet-0f82985187f3f7b70
Private_Subnet_1A

VPC: vpc-03279cdb550dee663 Owner: 463646775279
Availability Zone: ap-south-1a (aps1-az1) Zone type: Availability Zone
IP addresses available: 59 CIDR: 192.168.0.128/26

Create new subnet ↗

Auto-assign public IP | [Info](#)
Disable

- Now select created VPC.
- And then Select subnet private_Subnet_1A.
- And Auto-assign public IP Disable.
- Scroll down.

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group

☒ Select existing security group

Common security groups [Info](#)

Select security groups ▼



web-SG sg-0f4faf3ca638a151e ✕
VPC: vpc-03279cdb550dee663

[Compare security group rules](#)

Security groups that you add or remove here will be added to or removed from all your network interfaces.

► Advanced network configuration

- Select Firewall (security group) existing security group.
- And select your created web-SG security group.
- Now we have our EC2 instance in a private subnet okay.
- Now how we will place our website over there because if you want to access this we have to create a EC2 instance connect End point in order to access this.
- Because our EC2 instance is in a private subnet and we don't have public IP.

▼ Advanced details [Info](#)

Domain join directory [Info](#)

Select ▼

[Create new directory](#)

IAM instance profile [Info](#)

Select ▼

[Create new IAM profile](#)

Hostname type [Info](#)

IP name ▼

DNS Hostname [Info](#)

- So, for that scroll down and open Advanced details.
- And scroll down.

User data - optional [Info](#)

Upload a file with your user data or enter it in the field.

[Choose file](#)

```
#!/bin/bash
yum update -y

# Install Apache
yum install -y httpd

# Start and enable Apache to run on boot
systemctl start httpd
systemctl enable httpd

# Create a simple HTML file in Root Directory
echo "This is the CloudFolks HUB Page served at Root" >
/var/www/html/index.html
```

- The script paste in Advanced details section User data.
- Script is provided in Resource.
- Now simply scroll down.

Software Image (AMI)

Amazon Linux 2023 AMI 2023.11.....[read more](#)

ami-0cc96c4cd98401dae

[Cancel](#)

[Launch instance](#)

[Preview code](#)

- Now leave remaining setting as it is.
- And simply scroll down and click Launch instance.
- Now this instance is ready now we need to create similar two instance with name cloudfolk.in/aws and cloudfolk.in/azure.
- And the both instance script is provided in Resource.

EC2 > Instances

Instances (4) [Info](#) Last updated 3 minutes ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Saved filter sets Choose filter set Find Instance by attribute or tag (case-sensitive) < 1 > ⚙

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm st
☐	Instance1	i-0b0c4a93e23a844bb	Running	t2.medium	2/2 checks passec	View alai
☐	cloudfolk.in/aws	i-01724801d11c35a3a	Running	t3.micro	3/3 checks passec	View alai
☐	cloudfolk.in	i-0922d7e2b43dd65ed	Running	t3.micro	3/3 checks passec	View alai
☐	cloudfolk.in/a...	i-01fb097c28bb5777a	Running	t3.micro	3/3 checks passec	View alai

- Now our three instance is created.
- Now we need to create three target group.
- Simply click Target group in Load Balancing section inside nav bar.

Target groups (3) [Info](#) | [What's new?](#)

[Actions](#) ▼[Create target group](#)

< 1 >



Name



ARN



Port



Protocol

- Now simply click Create target group.

Create target group

A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health checks on the targets.

Settings - immutable

Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation."

Target type

Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

☒ Instances

Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.

Suitable for: ALB NLB GWLB

☐ IP addresses

Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.

Suitable for: ALB NLB GWLB

☐ Lambda function

Supports load balancing to a single Lambda function. ALB required as traffic source.

Suitable for: ALB

☐ Application Load Balancer

Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.

Suitable for: NLB

- Now select setting Instances.

Target group name

Name must be unique per Region per AWS account.

Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 9/32

Protocol

Protocol for communication between the load balancer and targets.



Port

Port number where targets receive traffic. Can be overridden for individual targets during registration.



1-65535

- Now enter target group name.
- And scroll down.

IP address type

Only targets with the indicated IP address type can be registered to this target group.

☒ IPv4

Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.

☐ IPv6

Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#) 

VPC

Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

vpc-03279cdb550dee663 (MyVPC)
192.168.0.0/24



- Now select your created VPC.
- Now simply scroll down.

Health checks

The associated load balancer periodically sends requests, per the settings below, to the registered targets to test their status.

Health check protocol

HTTP



Health check path

Use the default path of "/" to perform health checks on the root, or specify a custom path if preferred.

/

Up to 1024 characters allowed.

- In health checks path don't do any thing in this target group only.

► Tags - optional

Consider adding tags to your target group. Tags enable you to categorize your AWS resources so you can more easily manage them.

Cancel

Next

- And click Next.

Available instances (1/3)

Filter instances

	Instance ID	Name	State	Security groups
☐	i-01fb097c28bb5777a	cloudfolk.in/azure	Running	web-SG
☐	i-01724801d11c35a3a	cloudfolk.in/aws	Running	web-SG
☒	i-0922d7e2b43dd65ed	cloudfolk.in	Running	web-SG

1 selected

Ports for the selected instances
Ports for routing traffic to the selected instances.

80

1-65535 (separate multiple ports with commas)

[Include as pending below](#)

- Now select cloudfolk.in instance for this target cloudfolk.
- And then click include as pending below.

Review targets

Targets (1) [Remove all pending](#)

Filter targets ☐ Show only pending

Instance ID	Name	Port	State	Security groups	Zone	Private IPv4 address	Subnet
i-0922d7e2b43dd65ed	cloudfolk.in	80	Running	web-SG	ap-south-1a	192.168.0.140	subn

1 pending

[Cancel](#) [Previous](#) [Next](#)

- Now simply click Next.

Step 2: Register targets [Edit](#)

Targets (1)

Instance ID	Name	Port	Zone
i-0922d7e2b43dd65ed	cloudfolk.in	80	ap-south-1a

[Cancel](#) [Previous](#) [Create target group](#)

- Now in next step scroll and click Create target group.
- Our cloudfolk target group is created.
- Now we need to create two more target group aws and azure.
- Similar steps but only two changes in health check path in aws give /aws/, and in azure target group enter /azure/.
- And second change is in aws target group select cloudfolk.in/aws instance and in azure target group select cloudfolk.in/azure.

Target groups (3) [Info](#) | [What's new?](#)

[Actions](#)[Create target group](#)

< 1 >



☐	Name	ARN	Port	Protocol	Target type
☐	aws	arn:aws:elasticloadbalancin...	80	HTTP	Instance
☐	azure	arn:aws:elasticloadbalancin...	80	HTTP	Instance
☐	cloudfolk	arn:aws:elasticloadbalancin...	80	HTTP	Instance

- Now three target group is created.
- Now we are going to create a Load Balancer.
- Click Load Balancer.

Load balancers [What's new?](#)

[Actions](#)[Create load balancer](#)

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

< 1 >



☐	Name	State	Type	Scheme	IP address type	V
--------------------------	------	-------	------	--------	-----------------	---

- Now simply click Create load balancer.

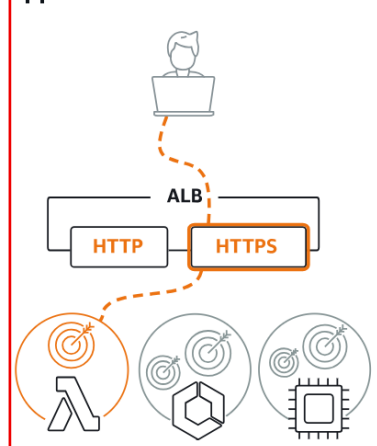
[EC2](#) > [Load balancers](#) > Compare and select load balancer type

Compare and select load balancer type

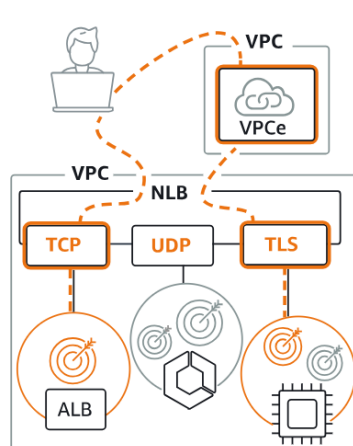
A complete feature-by-feature comparison along with detailed highlights is also available. [Learn more](#)

Load balancer types

Application Load Balancer [Info](#)



Network Load Balancer [Info](#)



Gateway Load Balancer [Info](#)



- Now we are going to create Application Load Balancer.
- Scroll down and click create.

Basic configuration

Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

Alb

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme [Info](#)

Scheme can't be changed after the load balancer is created.

☒ Internet-facing

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☐ Internal

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.
- Compatible with the IPv4 and Dualstack IP address types.

- Now Enter load balancer name.
- Leave default everything and scroll down.

Network mapping [Info](#)

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC [Info](#)

The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#).

vpc-03279cdb550dee663 (MyVPC)
192.168.0.0/24



[Create VPC](#)

IP pools [Info](#)

You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view Pools in the [Amazon VPC IP Address Manager console](#).

☐ Use IPAM pool for public IPv4 addresses

The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

- Now select your created VPC.

Availability Zones and subnets [Info](#)

Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☒ ap-south-1a (aps1-az1)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and internet gateway (IGW) routes configured to allow traffic.

subnet-0e7823bb0276394a2 (Public_Subnet_1A)
192.168.0.0/26
Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (✓)

☒ ap-south-1b (aps1-az3)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and internet gateway (IGW) routes configured to allow traffic.

subnet-0eeb1e9dd70507728 (Public_Subnet_2B)
192.168.0.64/26
Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (✓)

- Now select Availability Zones and subnets (Public_Subnet_1A & Public_Subnet_2B).

Security groups [Info](#)

A security group is a set of firewall rules that control the traffic to your load balancer. Listed security groups are in the same VPC as you selected above. Selected security groups must include at least 1 inbound rule and 1 outbound rule to allow traffic to your load balancer.

Security groups

Select up to 5 security groups

Alb-SG (sg-0b3ee61463ef94613) ✕
Inbound rule (1) Outbound rule (1)



[Create security group](#)

- Now Select security group Alb-SG.
- And scroll down.

Default action [Info](#)

The default action is used if no other rules apply. Choose the default action for traffic on this listener.

Routing action

☒ Forward to target groups

☐ Redirect to URL

☐ Return fixed response

Forward to target group [Info](#)

Choose a target group and specify routing weight or [create target group](#).

Target group

cloudfolk
Target type: Instance, IPv4 | Target stickiness: Off

HTTP



Weight

1

0-999

Percent

100%

- Now in default action select target group cloudfolk only now.
- Now scroll down.

Creation workflow and status

► Server-side tasks and status

After completing and submitting the above steps, all server-side tasks and their statuses become available for monitoring.

[Cancel](#)

[Create load balancer](#)

- Now simply click Create load balancer.
- One thing in adding target group add only cloudfolk target group.

Load balancers (1) [What's new?](#)



[Actions](#)

[Create load balancer](#)

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

< 1 >



☐	Name	State	Type	Scheme	IP address type	VPC
☐	Alb	Provisioning..	application	Internet-facing	IPv4	vpc

- Load balancer is created wait for state ready.
- And check your attach target group is health status is healthy or not.

Targets
Monitoring
Health checks
Attributes
Tags

Registered targets (1) [Info](#)
Anomaly mitigation: Not applicable
Deregister
Register targets

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

Filter targets
1

☐	Instance ID	Name	Port	Zone	Health status	Health s
☐	i-0922d7e2b43dd65ed	cloudfolk.in	80	ap-south-1a (a...	Initial	Target re

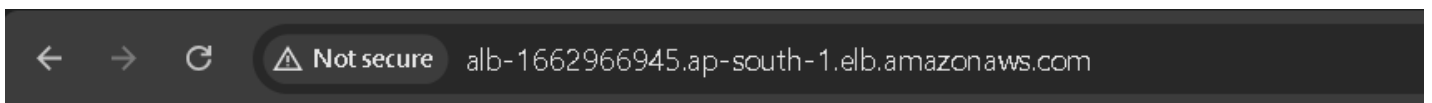
- Go in target group and open cloudfolk and scroll down to register target.
- And check health status is healthy or not.
- When all thing done load balancer is Active and target group is healthy open created load balancer and copy dns and open in new tab.

Alb
Actions

▼ Details

Load balancer type Application	Status Active	VPC vpc-03279cdb550dee663	Load balancer IP address type IPv4
Scheme Internet-facing	Hosted zone ZP97RAFLXTNZK	Availability Zones subnet-0e7823bb0276394a2 ap-south-1a (aps1-az1) subnet-0eeb1e9dd70507728 ap-south-1b (aps1-az3)	Date created May 13, 2026, 17:43 (UTC+05:30)
Load balancer ARN arn:aws:elasticloadbalancing:ap-south-1:463646775279:loadbalancer/app/Alb/717e8fecf98d3143		DNS name Info Alb-1662966945.ap-south-1.elb.amazonaws.com (A Record)	

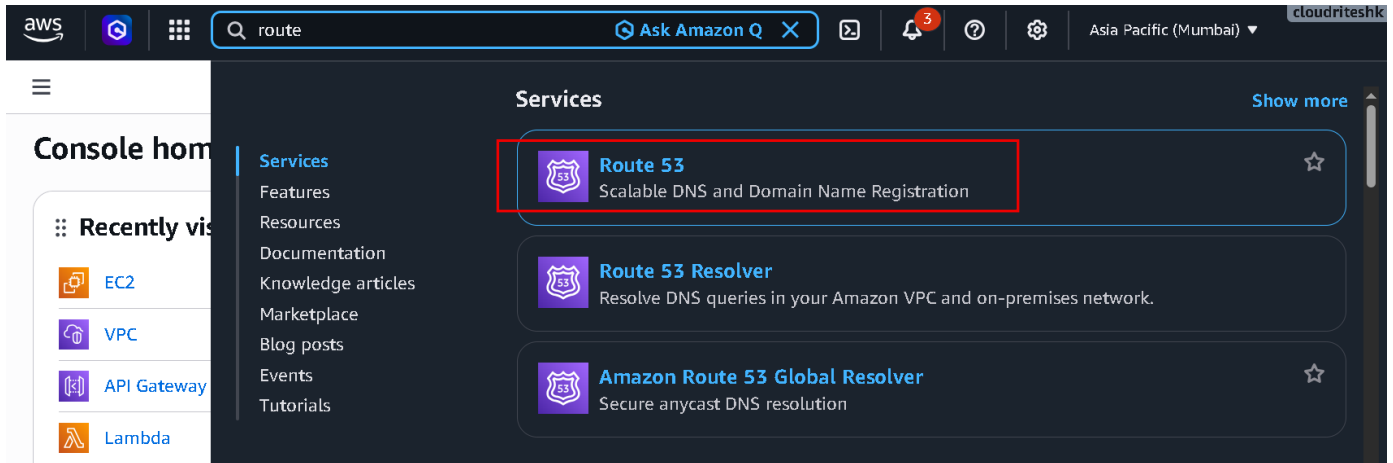
- Copy this DNS name.



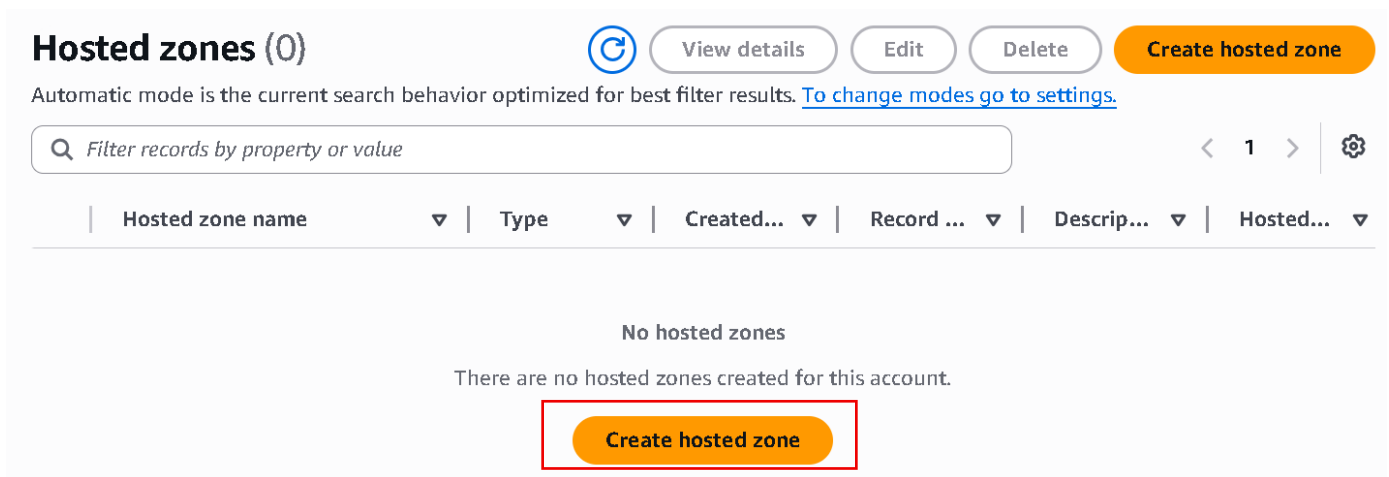
This is the CloudFolks HUB Page served at Root

- It's working.

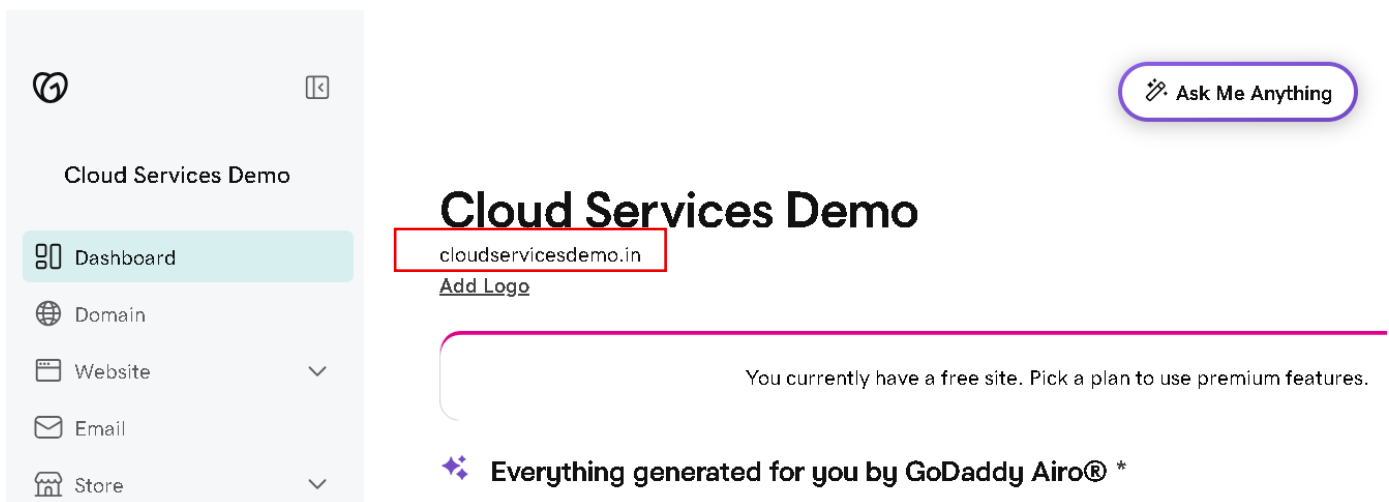
- Now one thing in our real world Scenario we have no such type of URL this URL are provided by AWS.
- Now we solve this problem.



- Open Aws in new tab and search route 53.
- And open it.



- Click Create hosted Zone.
- For creating hosted zone you need to buy a domain name.
- I purchased my domain name from GoDaddy.
- Go to GoDaddy and copy the domain name.



- My domain name is cloudservicedemo.in

Create hosted zone [Info](#)

Hosted zone configuration

A hosted zone is a container that holds information about how you want to route traffic for a domain, such as example.com, and its subdomains.

Domain name [Info](#)

This is the name of the domain that you want to route traffic for.

Valid characters: a-z, 0-9, ! " # \$ % & ' () * + , - / : ; < = > ? @ [\] ^ _ ` { | } . ~

- Enter Domain name.
- And simply scroll down.

Tags [Info](#)

Apply tags to hosted zones to help organize and identify them.

No tags associated with the resource.

[Add tag](#)

You can add up to 50 more tags.

[Cancel](#)[Create hosted zone](#)

- Click Create hosted zone.

[Public](#) **cloudservicesdemo.in** [Info](#)[Delete zone](#)[Test record](#)[Configure query logging](#)

► Hosted zone details

[Edit hosted zone](#)[Records \(2\)](#)[Accelerated recovery](#)[DNSSEC signing](#)[Hosted zone tags \(0\)](#)

Records (2) [Info](#)

[Delete record](#)[Import zone file](#)[Create record](#)

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

- Now simply click Create record.

Create record [Info](#)

Quick create record

[Switch to wizard](#)

▼ Record 1

[Delete](#)

Record name [Info](#)

.cloudservicesdemo.in

Keep blank to create a record for the root domain.

☒ Alias

Record type [Info](#)

A – Routes traffic to an IPv4 address and some AWS resources

- Record name as learn.
- And on the Alias.

Alias

Route traffic to [Info](#)

Alias to Application and Classic Load Balancer

Asia Pacific (Mumbai)

Q dualstack.Alb-1234617448.ap-south-1.elb.amazonaws.com

Alias hosted zone ID: ZP97RAFLXTNZK

Routing policy [Info](#)

Simple routing

Evaluate target health ☒ Yes

[Add another record](#)

[Cancel](#) [Create records](#)

- Now select Alias to Application and classic Load Balancer.
- Asia-Pacific (Mumbai)
- And choose the created load balancer.
- And simply click Create records.
- Now records are created.
- But before this go to GoDaddy and add this in DNS.

Public **cloudservicesdemo.in** [Info](#) [Delete zone](#) [Test record](#) [Configure query logging](#)

▼ **Hosted zone details** [Edit hosted zone](#)

Hosted zone name cloudservicesdemo.in	Query log -	Name servers ns-284.awsdns-35.com ns-728.awsdns-27.net ns-1823.awsdns-35.co.uk ns-1364.awsdns-42.org
Hosted zone ID Z0892389LU8VATY9TSUV	Type Public hosted zone	
Description -	Record count 2	

- These name we need to Enter in our Domain.
- Go to GoDaddy.

Cloud Services Demo

- Dashboard
- Domain**
- Website
- Email
- Store
- Appointments

Domain

cloudservicesdemo.in [Connect Domain](#)

Domain Status: IDLE

Overview **DNS** Registration Settings Products Activity Log

DNS Records Forwarding **Nameservers** Hostnames DS Records

Nameservers determine where your DNS is hosted and where you add, edit or delete your DNS records.

Using custom nameservers [Change Nameservers](#)

- Now go to Domain and then go inside DNS.
- And then click Change Nameservers.
- And one by one add name servers.
- Open new tab in browser and enter learn.cloudservicedemo.in

This is the CloudFolks HUB Page served at Root

- Its working.
- Now our cloudfolk.in is opened but we want to also open cloudfolk.in/aws and cloudfolk.in/azure
- So for this go to your load balancer tab and open created load balancer.

Listeners and rules (1/1) [Info](#)

A listener checks for connection requests on its configured protocol and port. Requests received by the listener is routed according to the default action and any additional rules.

☒ Protocol:Port ☐ Default action ☐ Rules ☐ ARN ☐ Security policy

Protocol:Port	Default action	Rules	ARN	Security policy
☒ HTTP:80	Forward to target group cloudfolk 1 (100%) Target group stickiness: Off	1 rule	ARN	Not applicable

- Now go to the load balancer and open your created load balancer.
- And scroll down in Listeners and rules Select that one rule shows and then click Manage rules and select rule.

▼ Name and tags [Info](#)

Tags can help you manage, identify, organize, search for and filter resources.

Name

[Add additional tags](#)

- Now give name R1.

Conditions (0 values) [Info](#)

[Rule limits](#)

Define 1-5 condition values. Additional conditions can't be added once the limit is reached.

[Add condition](#) ▲

Host header

Path

Query string

HTTP request method

HTTP header

Source IP

- In condition select Host header.

Conditions (1 value) [Info](#)

[Rule limits](#)

Define 1-5 condition values. Additional conditions can't be added once the limit is reached.

▼ Host header (value) =

[Remove](#)

Match pattern type

☒ Value matching
Match with glob syntax, using `*` and `?` as wildcards.

☐ Regex matching
Match with regex syntax.

Host header condition value

Valid domain name according to DNS standards. For example: `*.example.com`

=

Valid characters are a-z, A-Z, 0-9 and [special characters](#). Host header must be 1-128 characters. Character count: 24/128

[+ Add OR condition value](#)

- Now Enter aws.cloudservicesdemo.in (this is my domain name cloudservicesdemo.in)
- Scroll down.

Actions [Info](#)

Requests matching all rule conditions route according to the rule actions.

Routing action

☒ Forward to target groups

☐ Redirect to URL

☐ Return fixed response

Forward to target group [Info](#)

Choose a target group and specify routing weight or [create target group](#).

Target group

Target type: Instance, IPv4 | Target stickiness: Off

HTTP ▼

**Weight**

0-999

Percent

100%

- Now we need to select our aws target group in Action.

Listener rules (2) [Info](#) [Rule limits](#) [Reset priorities](#) [Add gap between priorities](#) ▼

Traffic received by the listener is routed according to the default action and any additional rules. Rules are evaluated in priority order from the lowest value to the highest value.

	Priority ✎	Name tag	Conditions (If)	Transforms
⋮	100 Priority value must be 1-50,000.	R1	Path (value) = <code>/aws*</code>	-
	Last (default)	Default	If no other rule applies	-

[Cancel](#) [Previous](#) [Next](#)

- Now enter the priority for you R1.
- And simply click next.
- Now in next step simply scroll down and click add rules.
- Now rules are added.
- Go to target group and open aws target group.

Targets [Monitoring](#) [Health checks](#) [Attributes](#) [Tags](#)

Registered targets (1) [Info](#) [Anomaly mitigation: Not applicable](#) [Deregister](#) [Register targets](#)

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

☐	Instance ID	Name	Port	Zone	Health status	Health s
☐	i-01724801d11c35a3a	cloudfolk.in/aws	80	ap-south-1a (a...	✓ Healthy	-

- Now scroll down and check our target health status.
- Now go to route 53 in new tab.
- And create new record.

Create record [Info](#)

Quick create record [Switch to wizard](#)

▼ Record 1 [Delete](#)

Record name [Info](#)

aws .cloudservicesdemo.in

Keep blank to create a record for the root domain.

Record type [Info](#)

A – Routes traffic to an IPv4 address and some AWS resources ▼

☒ Alias

- Enter aws.
- And open alias.

Alias

Route traffic to [Info](#)

Alias to Application and Classic Load Balancer

Asia Pacific (Mumbai)

Q dualstack.Alb-1662966945.ap-south-1.elb.amazonaws.com

Alias hosted zone ID: ZP97RAFLXTNZK

Routing policy [Info](#)

Simple routing

Evaluate target health

☒ Yes

- Fill these details like previously we done.
- And scroll down and click create record.

Records (4) [Info](#)



Delete record

Import zone file

Create record

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Q Filter records by property or value

Type

Routing p...

Alias

< 1 >



☐	Record name	Type	Routin...	Differ...	Alias	Value/Route traffic to	TTL (...	Health
☐	cloudservicesdemo.in	NS	Simple	-	No	ns-386.awsdns-48.com. ns-777.awsdns-33.net. ns-1183.awsdns-19.org. ns-1793.awsdns-32.co.uk.	172800	-
☐	cloudservicesdemo.in	SOA	Simple	-	No	ns-386.awsdns-48.com. aws...	900	-
☐	aws.cloudservicesdemo.in	A	Simple	-	Yes	dualstack.alb-1662966945.a...	-	-
☐	learn.cloudservicesdemo.in	A	Simple	-	Yes	dualstack.alb-1662966945.a...	-	-

- Now our record is created
- Copy aws.cloudservicesdemo.in and open in new tab.



It works!

- Now its working all thing completed
- Now for azure repeat same steps.